

Exercise XAI3: Partial Dependence Plots and Model Interpretability

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1 Introduction

This report analyses two regression models using Partial Dependence Plots (PDPs). The objective is to explain how selected input variables influence the predictions of Random Forest models in two contexts: bike rental demand and house price estimation.

Interpretability is the ability to understand how a model produces its predictions. PDPs are a global and model-agnostic method: they can be applied to any fitted model and describe average model behaviour over the dataset. A PDP shows the marginal effect of one feature, or a pair of features, on the prediction of an already trained model. This is an explanation of the fitted model, not proof of real-world causality.

2 Methodology

2.1 Dataset choice

For bike rental prediction, the daily dataset `DATA/day.csv` was used. The file contains 731 daily observations and directly matches the requested target `cnt` and the requested predictors `instant`, `temp`, `hum`, and `windspeed`. The hourly file `DATA/hour.csv` contains 17,379 observations and could support a different hourly demand analysis, but it would introduce hourly variation that is not part of the requested interpretation of days since 2011. Therefore, `day.csv` is the more appropriate source for this analysis.

For house price prediction, the dataset `DATA/kc_house_data.csv` was used. The target variable is `price`, and the model uses `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, `floors`, and `yr_built`.

2.2 Model training

Only the required variables were selected, numeric conversion was applied where necessary, and missing values were removed. Each dataset was split into an 80% training partition and a 20% test partition using `set.seed(123)`. A Random Forest regression model was fitted on each training partition.

2.3 Evaluation metrics

Before interpreting a model with PDPs, it is important to check that its predictions are reasonably informative. If predictive quality were very poor, PDPs would still describe the fitted model, but the explanation would be less useful. The following metrics were computed on the 20% test partitions.

Dataset	RMSE	MAE	R^2
Bike rentals	732.87	563.25	0.868
House prices	225155.66	142073.50	0.665

Table 1: Random Forest regression performance on test partitions.

The bike rental model shows strong predictive behaviour on the test partition. The house price model is less accurate but still captures a meaningful part of the variation in prices. Therefore, PDP interpretation is useful for both models, with more caution for the house model.

2.4 PDP computation

One-dimensional PDPs were computed for the requested variables. A two-dimensional PDP was computed for temperature and humidity in the bike rental model. Sampling was used for PDP computation where appropriate to keep the analysis efficient, especially for the larger house price dataset.

3 Bike Rental Prediction

3.1 Model performance

The bike rental model obtained an RMSE of 732.87, an MAE of 563.25, and an R^2 of 0.868 on the test partition. This indicates that the model captures the main predictive structure of daily bike counts, so its PDPs provide a meaningful summary of model behaviour.

3.2 PDP: days since 2011

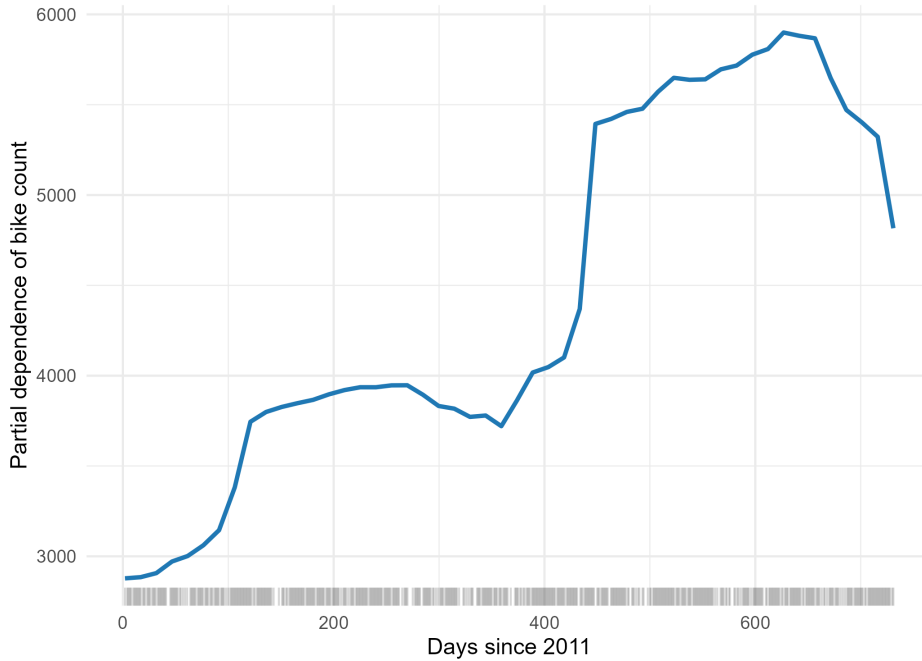


Figure 1: Partial dependence of days since 2011 for predicted bike rentals.

The curve is strongly non-linear. Predicted rentals increase from the beginning of the period, rise sharply around the middle of the timeline, and decrease near the final days. This means that the

Random Forest learned an important time-related pattern in the daily counts. Since `instant` is a time index, the interpretation should be treated as model behaviour over the observed period rather than as a causal effect of time itself.

3.3 PDP: temperature

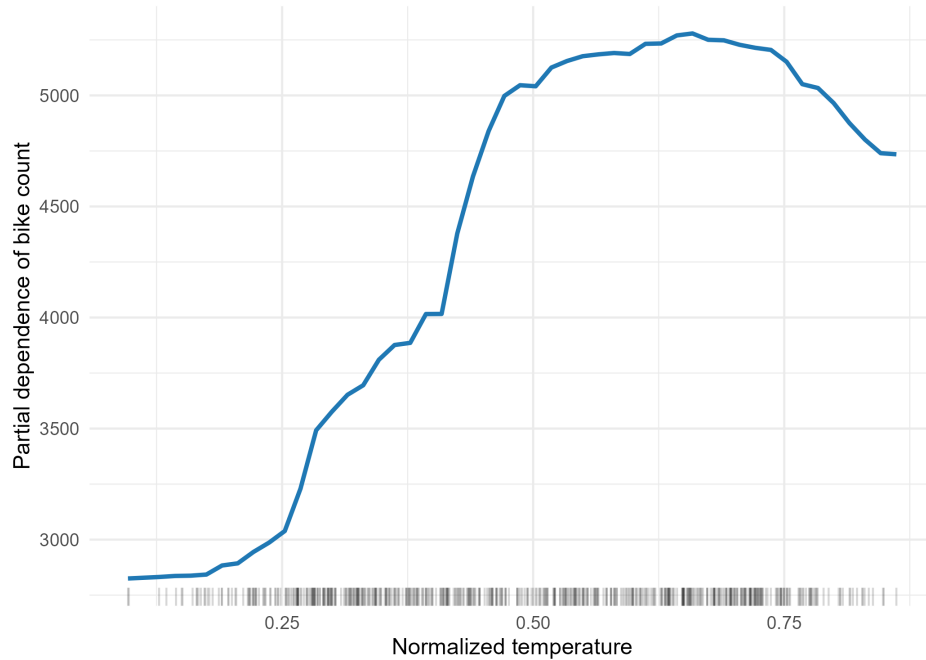


Figure 2: Partial dependence of temperature for predicted bike rentals.

Temperature has a clear non-linear effect. The model predicts low rental counts at low temperatures, then a strong increase up to warm conditions. After the peak, the prediction decreases slightly at the highest temperatures. The rug marks show fewer observations at the extreme ends, so the lowest and highest temperature regions should be interpreted more cautiously.

3.4 PDP: humidity

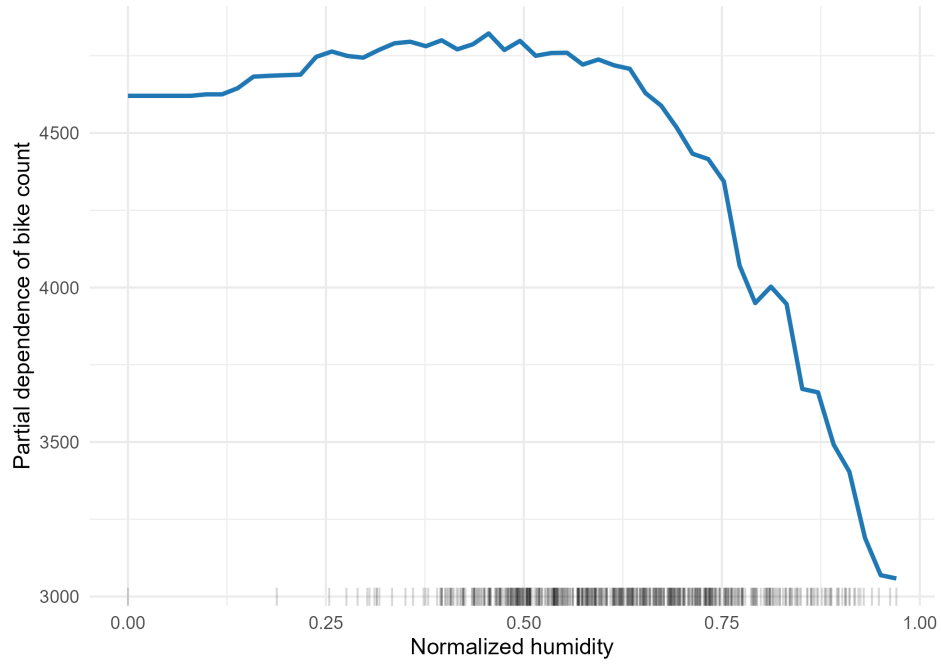


Figure 3: Partial dependence of humidity for predicted bike rentals.

Humidity is almost flat to slightly positive at low and medium values, but the curve decreases strongly when humidity becomes high. For the Random Forest model, very humid days are associated with lower predicted rental counts. The highest humidity values have less empirical support than the central range, so the exact shape at the far right should be interpreted carefully.

3.5 PDP: wind speed

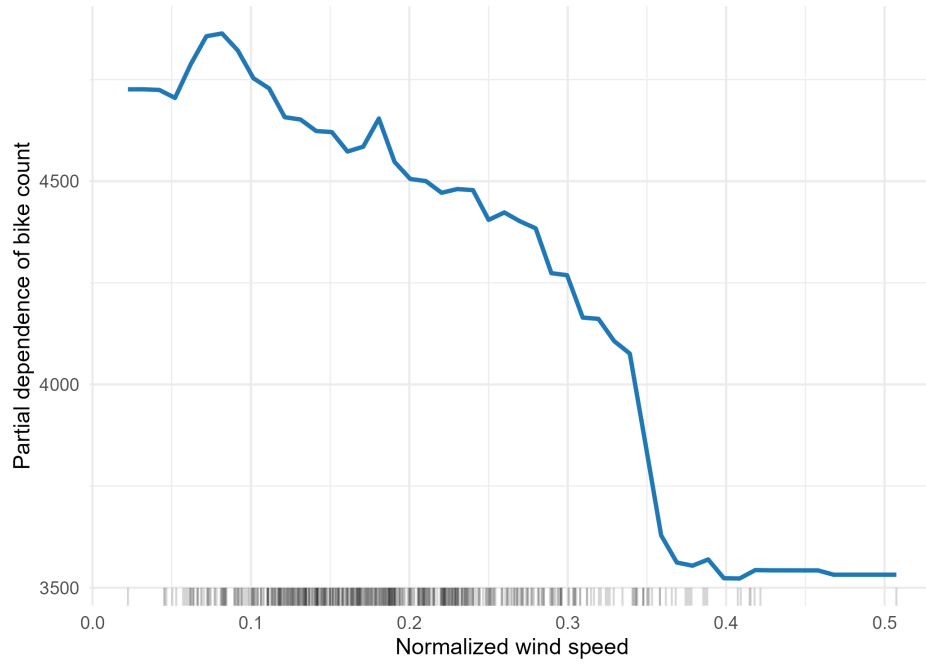


Figure 4: Partial dependence of wind speed for predicted bike rentals.

Wind speed has a mostly decreasing effect. The model predicts higher counts under low wind speed and progressively lower counts as wind speed increases, with a sharper drop after medium wind values. The right side of the plot is sparse, so the plateau at the highest wind speeds should be read with caution.

3.6 2D PDP: temperature and humidity

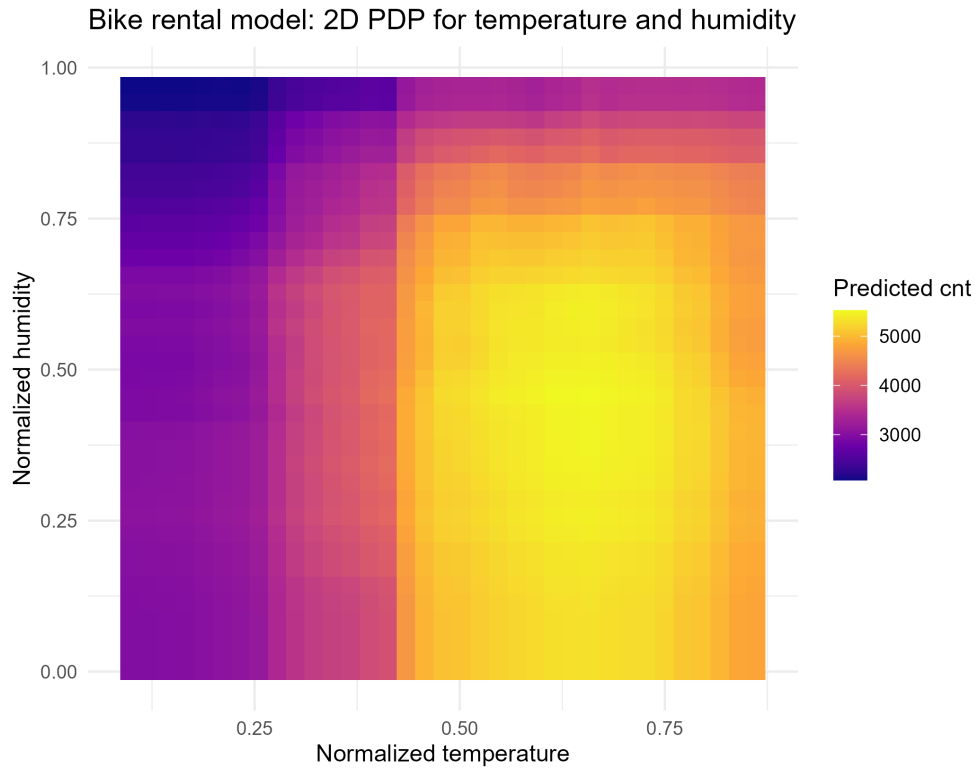


Figure 5: Two-dimensional PDP of temperature and humidity for predicted bike rentals.

The two-dimensional PDP shows that predicted rentals are highest at medium-high temperatures and low or moderate humidity. Cold temperatures produce lower predictions across most humidity values. At warm temperatures, high humidity reduces the predicted count. This indicates that the Random Forest model favours warm and relatively dry conditions.

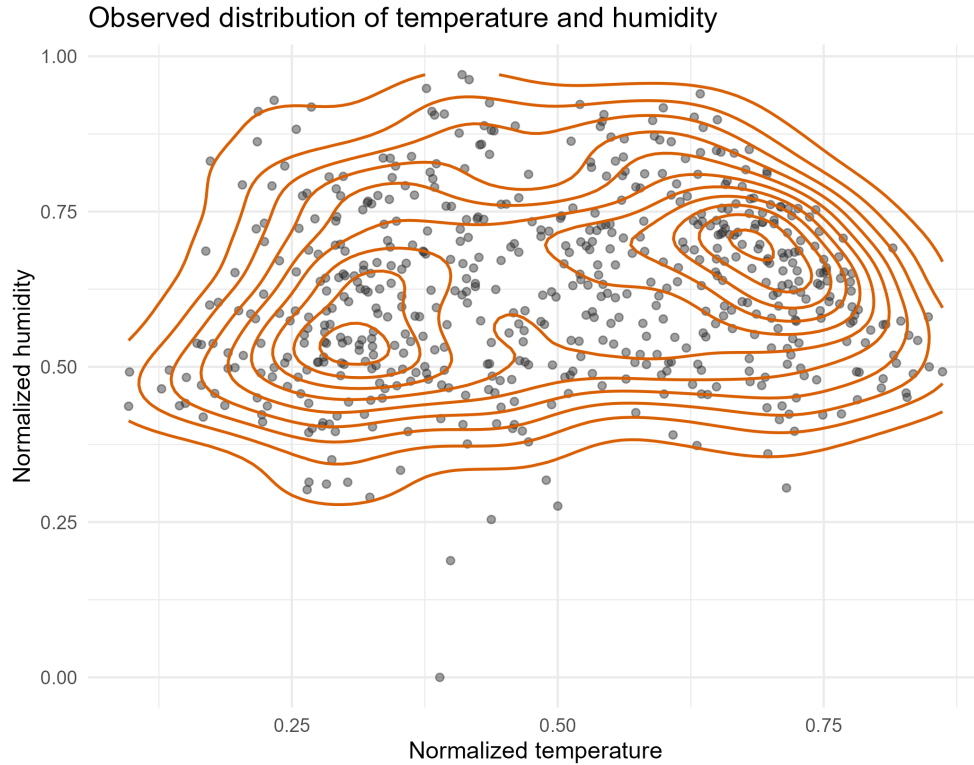


Figure 6: Observed distribution of temperature and humidity in the bike dataset.

The distribution plot shows where the observed data are concentrated. Many observations appear around moderate humidity and medium to high temperatures, while some extreme combinations are rare. This matters for the 2D PDP because regions with few observations are less reliable: the model is averaging predictions over feature combinations with limited empirical support.

3.7 Answer to the exercise question

For the bike rental model, days since 2011 have a strong non-linear influence: predicted counts increase during much of the observed period and fall near the end. Temperature also has a non-linear effect, increasing predictions up to warm values and decreasing slightly at the highest values. Humidity is mainly negative at high levels, while wind speed has a mostly decreasing effect. The joint temperature-humidity PDP shows that the model predicts the highest bike rentals on warm days with low or moderate humidity.

4 House Price Prediction

4.1 Model performance

The house price model obtained an RMSE of 225155.66, an MAE of 142073.50, and an R^2 of 0.665 on the test partition. This performance is moderate: the model captures relevant price structure, but the PDPs should be interpreted as broad model tendencies rather than precise pricing rules.

4.2 PDP: bedrooms

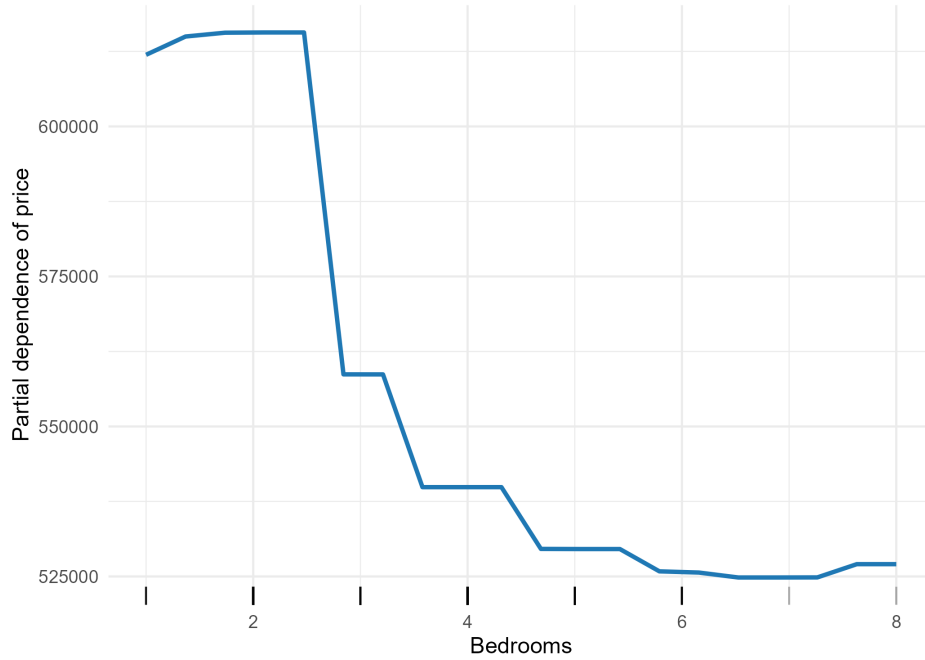


Figure 7: Partial dependence of bedrooms for predicted house price.

The bedrooms curve is non-linear and mostly decreasing after the first values. The model predicts slightly higher prices for houses with one to two bedrooms, then lower prices as bedroom count increases. This does not mean that bedrooms reduce real value causally; rather, after averaging over the other selected variables, the Random Forest does not use additional bedrooms as a simple positive price signal. High bedroom counts are relatively sparse, so the far-right part of the curve should be treated cautiously.

4.3 PDP: bathrooms

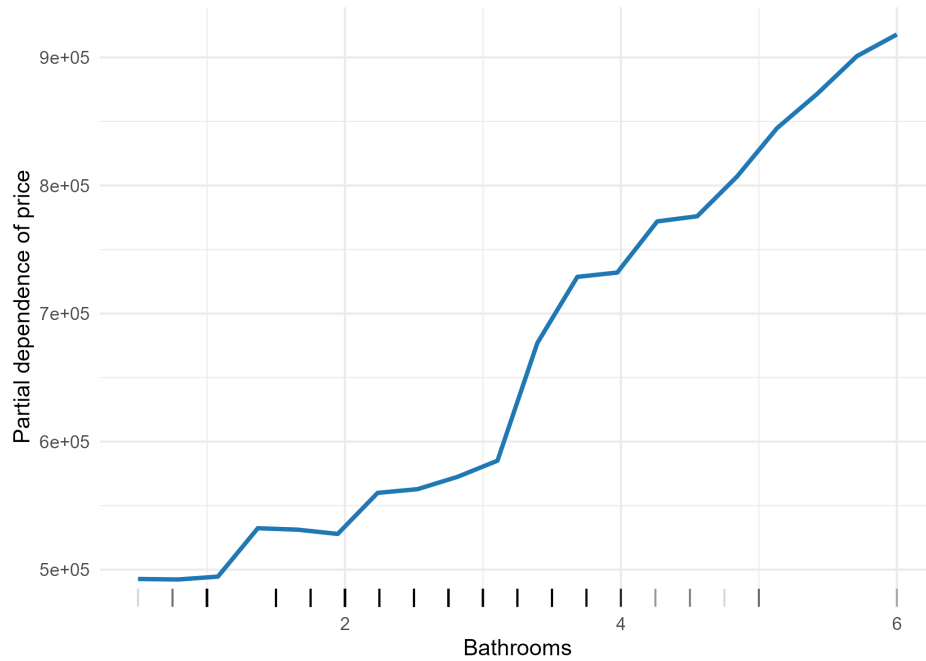


Figure 8: Partial dependence of bathrooms for predicted house price.

Bathrooms show a clear increasing effect. The predicted price rises slowly at low bathroom counts and then more strongly from around three bathrooms onward. This means the Random Forest associates more bathrooms with higher predicted prices. The highest bathroom counts have fewer observations, so the exact slope in that region is less reliable.

4.4 PDP: sqft_living

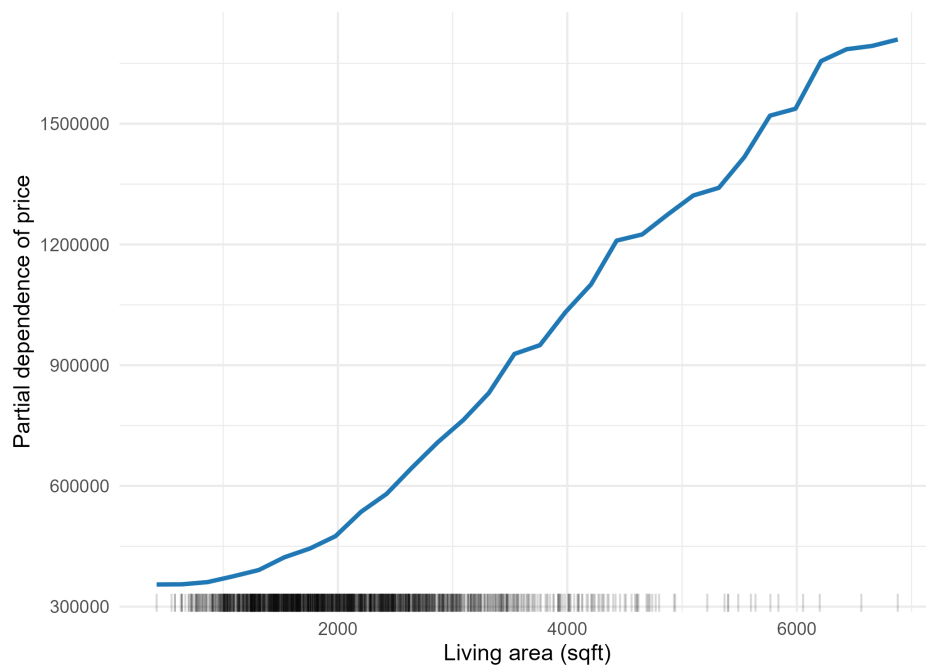


Figure 9: Partial dependence of living area for predicted house price.

The effect of `sqft_living` is strongly increasing. Predicted price rises steadily as living area grows, making this the clearest positive relationship in the house model. The rug marks show that very large houses are less common, so the largest living-area values should be interpreted with more caution than the central range.

4.5 PDP: floors

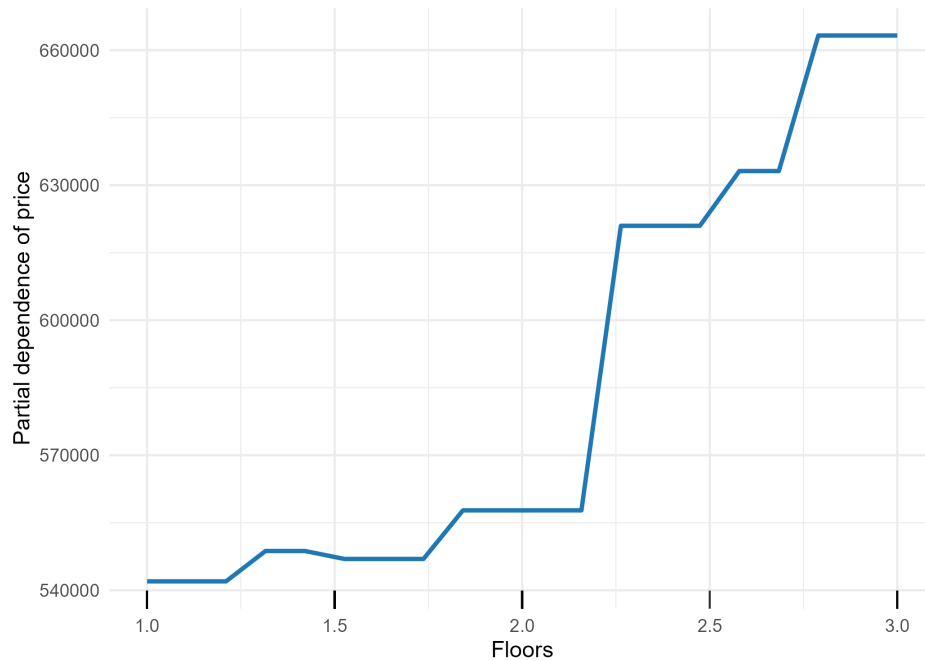


Figure 10: Partial dependence of floors for predicted house price.

Floors have a positive step-like effect. The curve is flat over intervals because floors take a limited number of values, then increases at higher floor counts. The Random Forest therefore associates more floors with higher predicted prices, although the effect is much smaller than the effect of living area. Values near the maximum number of floors are sparse and should be interpreted cautiously.

4.6 Answer to the exercise question

For the house price model, bedrooms show a non-linear pattern and do not produce a simple increasing effect. Bathrooms increase predicted price, especially in the medium-high range. Living area has the strongest positive effect: larger houses receive much higher predictions. Floors also increase predicted price, but in a smaller and step-like way.

5 Discussion

PDPs provide a global explanation of the fitted Random Forest models. They show average model behaviour, not individual behaviour for a specific observation. This is visible in the plots: each curve summarises the average predicted response after averaging over the rest of the dataset.

The main limitation is that PDPs assume that the analysed features are not strongly correlated with the remaining features. If this assumption is not reasonable, the PDP may evaluate combinations of variables that are uncommon or unrealistic. The temperature-humidity distribution plot illustrates another limitation: PDP values in sparse regions are less reliable because

they depend on feature combinations with limited empirical support. Finally, PDPs explain the fitted model and should not be interpreted as evidence of real-world causality.

6 Conclusions

The bike rental model has strong predictive quality and its PDPs show clear model behaviour: predicted rentals increase over much of the observed time period, rise with temperature until warm conditions, decrease at high humidity, and decrease with wind speed. The two-dimensional PDP indicates that warm and relatively dry conditions produce the highest predicted rental counts.

The house price model has moderate predictive quality. Its PDPs show that living area is the strongest positive driver of predicted price, bathrooms and floors also increase predictions, and bedrooms have a non-linear effect that is not simply positive.

Overall, PDPs provide a concise global interpretation of how the Random Forest models use the selected variables.